DATA STRUCTURE

ARRAYS:-

Q. 1D Array

```
#include <iostream>
using namespace std;

class Array
{
  int a[10], n;

public:
  void input()
  {
    cout << "Enter size: ";
    cin >> n;
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++)
    {
      cin >> a[i];
    }
  }
  void display()
  {
    cout << "Array elements: ";
    for(int i = 0; i < n; i++)
```

```
    {
        cout << a[i] << " ";
    }
  }
};

int main()
{
  Array obj;
  obj.input();
  obj.display();
  return 0;
}
```

Output:

```
Enter size: 5
Enter elements: 10 20 30 40 50
Array elements: 10 20 30 40 50
```

Q. 2D Array

```
#include <iostream>
using namespace std;

class Matrix
{
  int a[10][10], r, c;

public:
```

```
void input()
{
   cout << "Enter rows and columns: ";
   cin >> r >> c;

   cout << "Enter elements: ";
   for(int i = 0; i < r; i++)
   {
      for(int j = 0; j < c; j++)
      {
         cin >> a[i][j];
      }
   }
}

void display()
{
   cout << "Matrix is:\n";

   for(int i = 0; i < r; i++)
   {
      for(int j = 0; j < c; j++)
      {
         cout << a[i][j] << " ";
      }
      cout << endl;
```

```
    }
  }
};

int main()
{
  Matrix obj;
  obj.input();
  obj.display();

  return 0;
}
```

OUTPUT:

```
Enter rows and columns: 2 4
Enter elements: 22 33 44  55 66 66 66
66
Matrix is:
22 33 44 55
66 66 66 66
```

Q. Insertion in array

```
#include <iostream>
using namespace std;

class Array
{
  int a[10], n;

public:
```

```
void input()
{
   cout << "Enter size: ";
   cin >> n;

   cout << "Enter elements: ";
   for(int i = 0; i < n; i++)
   {
      cin >> a[i];
   }
}

void insert()
{
   int pos, val;

   cout << "Enter position: ";
   cin >> pos;

   cout << "Enter value: ";
   cin >> val;

   if(pos < 1 || pos > n + 1 || n >= 10)
   {
      cout << "Invalid position or array full";
      return;
```

```
        }

        for(int i = n; i >= pos; i--)
        {
            a[i] = a[i - 1];
        }

        a[pos - 1] = val;
        n++;
    }

    void display()
    {
        cout << "Array: ";
        for(int i = 0; i < n; i++)
        {
            cout << a[i] << " ";
        }
    }
};

int main()
{
    Array obj;
    obj.input();
    obj.insert();
```

```
    obj.display();


    return 0;
}
```

OUTPUT:

```
Enter size: 4
Enter elements: 1 2 3 4
Enter position: 2
Enter value: 5
Array: 1 5 2 3 4
```

Q. Deletion in array

```
#include <iostream>

using namespace std;


class Array
{
    int a[10], n;


public:
    void input()
    {
        cout << "Enter size: ";
        cin >> n;


        cout << "Enter elements: ";
        for(int i = 0; i < n; i++)
        {
```

```
        cin >> a[i];
    }
}

void deletion()
{
    int pos;

    cout << "Enter position to delete: ";
    cin >> pos;

    if(pos < 1 || pos > n)
    {
        cout << "Invalid position";
        return;
    }

    for(int i = pos - 1; i < n - 1; i++)
    {
        a[i] = a[i + 1];
    }

    n--;
}

void display()
```

```
  {
     cout << "Array: ";

     for(int i = 0; i < n; i++)
     {
        cout << a[i] << " ";
     }
  }
};

int main()
{
  Array obj;
  obj.input();
  obj.deletion();
  obj.display();

  return 0;
}
```

OUTPUT:

main.c... Output

```
Enter size: 5
Enter elements: 1 2 34  5 4
Enter position to delete: 3
Array: 1 2 5 4
```

Q. Matrix multiplication

```
#include <iostream>
using namespace std;

class Matrix
{
  int a[10][10], b[10][10], c[10][10];
  int r1, c1, r2, c2;

public:

  void input()
  {
    cout << "Enter rows and columns of first matrix: ";
    cin >> r1 >> c1;

    cout << "Enter elements of first matrix: ";

    for(int i = 0; i < r1; i++)
    {
      for(int j = 0; j < c1; j++)
      {
        cin >> a[i][j];
      }
    }
```

```
   cout << "Enter rows and columns of second matrix: ";
   cin >> r2 >> c2;

   cout << "Enter elements of second matrix: ";

   for(int i = 0; i < r2; i++)
   {
      for(int j = 0; j < c2; j++)
      {
         cin >> b[i][j];
      }
   }
}

void multiply()
{
   if(c1 != r2)
   {
      cout << "Matrix multiplication not possible";
      return;
   }

   for(int i = 0; i < r1; i++)
   {
      for(int j = 0; j < c2; j++)
      {
```

```
            c[i][j] = 0;

            for(int k = 0; k < c1; k++)
            {
               c[i][j] += a[i][k] * b[k][j];
            }
         }
      }
   }

   void display()
   {
      if(c1 != r2)
         return;

      cout << "Resultant Matrix:\n";

      for(int i = 0; i < r1; i++)
      {
         for(int j = 0; j < c2; j++)
         {
            cout << c[i][j] << " ";
         }
         cout << endl;
      }
   }
```

```
};

int main()
{
   Matrix obj;

   obj.input();
   obj.multiply();
   obj.display();

   return 0;
}
```

OUTPUT:

```
Enter rows and columns of first matrix: 2 2
Enter elements of first matrix: 1 2 3 4
Enter rows and columns of second matrix: 2 2
Enter elements of second matrix: 4 5 6 7
Resultant Matrix:
16 19
36 43
```